



Assignment

1

Solve the given equations:

① $\frac{3}{2}x - 4y = 6$

$2x + 3y = 8$

Solution

Let $\frac{3}{2}x - 4y = 6$ — (i)

$2x + 3y = 8$ — (ii)

$y = \frac{8 - 2x}{3}$ — (iii)

Putting the value of y in — (i) we get,

$$\frac{3x}{2} - \frac{3(8 - 2x)}{3} = 6$$

$9x - 64 + 16x = 6$

$9x + 16x = 36 + 64$

$25x = 100$

$\therefore \boxed{x = 4}$

Putting the value of x in — (iii) we get

$y = \frac{8 - 2 \times 4}{3}$

$y = \frac{0}{3}$

$\therefore \boxed{y = 0}$



② $3x^2 - 2x - 8 = 0$

Solution,

$$3x^2 - 2x - 8$$

Comparing it with to get a, b, c for $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Here,

$$a = 3$$

$$b = -2$$

$$c = -8$$

Now,

$$x = \frac{-(-2) \pm \sqrt{(-2)^2 - 4(3)(-8)}}{2(3)}$$

$$= \frac{-(-2) \pm \sqrt{4 + 96}}{2(3)}$$

$$= \frac{2 \pm 10}{6}$$

Here,

$$\text{For } x = \frac{2+10}{2}$$

$$\text{or } x = \frac{2-10}{2}$$

$$= \frac{12}{6}$$

$$= \frac{-8}{6}$$

$$\therefore x = 2$$

$$\text{or } x = -\frac{4}{3}$$



③ Find the value of x when

$$\left(\frac{3}{2}\right)^{-4} \div \left(\frac{2}{3}\right)^6 = \left(\frac{4}{9}\right)^{x-2}$$

Solution,

Given

$$\left(\frac{3}{2}\right)^{-4} \div \left(\frac{2}{3}\right)^6 = \left(\frac{4}{9}\right)^{x-2}$$

$$\therefore \left(\frac{3}{2}\right)^{-4} \div \left(\frac{2}{3}\right)^{-6} = \left(\frac{4}{9}\right)^{x-2} \quad \left(\because \text{same base in division then power is subtracted} \right)$$

$$\therefore \left(\frac{3}{2}\right)^{-4-(-6)} = \left(\frac{4}{9}\right)^{x-2}$$

$$\therefore \left(\frac{3}{2}\right)^2 = \left(\frac{4}{9}\right)^{x-2}$$

$$\therefore \left(\frac{3}{2}\right)^2 = \left(\frac{2}{3}\right)^{2x-4}$$

$$\therefore \left(\frac{2}{3}\right)^{-2} = \left(\frac{2}{3}\right)^{2x-4}$$

Since the base are same we can compare.

$$\therefore -2 = 2x-4$$

$$\therefore -2+4 = 2x$$

$$\therefore 2 = 2x$$

$$\therefore \boxed{x=1}$$



Assignment 2

$$\textcircled{1} f(x) = \begin{cases} a-bx & x > 2 \\ 5x & x = 2 \\ ax+b & x < 2 \end{cases}$$

if the function is continuous at $x=2$, find the value of a and b .

Solution,

$$f(x) = \begin{cases} a-bx, & x > 2 \\ 5x, & x = 2 \\ ax+b, & x < 2 \end{cases}$$

$f(x)$ is continuous at $x=2$

Now,

$$f(2) = 5 \times 2 \\ = 10$$

$$\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} (a-bx) = a-2b$$

$$\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} (ax+b) = 2a+b$$

Let the limit equal to each other and $f(2)$ to find a system of equation for a and b .

$$a - 2b = 10 \quad \text{--- (I)}$$

$$2a - b = 10 \quad \text{--- (II)}$$

Now

$$\text{eqn (I)}$$

$$2a + b = 10$$

$$b = 10 - 2a$$

Now,

$$b = 10 - 2a \text{ putting eqn --- (I)}$$

$$a - 2(10 - 2a) = 10$$

$$a - 20 + 4a = 10$$

$$5a = 10 + 20$$

$$5a = 30$$

$$\therefore a = \frac{30}{5}$$

$$\therefore a = 6$$

also,

$$b = 10 - 2a$$

$$b = 10 - 2 \times 6$$

$$= 10 - 12$$

$$= -2$$

$$\therefore a = 6, b = -2$$



② If $y = \frac{e^x \sin x}{2x^2 - 4x}$ find $\frac{dy}{dx}$

Solution

Given, $y = \frac{e^x \sin x}{2x^2 - 4x}$

we use quotient rule

$$\frac{d\left(\frac{u}{v}\right)}{dx} = \frac{u'v - uv'}{v^2}$$

Now,

$$u = e^x \sin x \therefore u' = e^x \sin x + e^x \cos x$$

$$v = 2x^2 - 4x \therefore v' = 4x - 4$$

Applying the rule we get

$$\therefore \frac{dy}{dx} = \frac{(e^x \sin x + e^x \cos x)(2x^2 - 4x) - (e^x \sin x)(4x - 4)}{(2x^2 - 4x)^2}$$

$$\therefore \frac{dy}{dx} = \frac{e^x [2x^2 (\sin x + \cos x) - 4x (\sin x + \cos x) - 4x \sin x + 4 \sin x]}{(2x^2 - 4x)^2}$$

$$\therefore \frac{dy}{dx} = \frac{e^x [2x^2 \sin x + 2x^2 \cos x - 4x \sin x - 4x \cos x - 4x \sin x + 4 \sin x]}{(2x^2 - 4x)^2}$$

$$\therefore \frac{dy}{dx} = \frac{e^x [2x^2 \sin x + 2x^2 \cos x - 8x \sin x - 4x \cos x + 4 \sin x]}{(2x^2 - 4x)^2}$$

$$\therefore \frac{dy}{dx} = \frac{e^x [\sin x (2x^2 - 8x + 4) + \cos x (2x^2 - 4x)]}{(2x^2 - 4x)^2}$$

$$\therefore \frac{dy}{dx} = \frac{2e^x [\sin x (x^2 - 4x + 2) + \cos x (x^2 - 2)]}{(2x^2 - 4x)^2}$$



Assignment +

3



① Solve:

$$2x - 3y + z = 4$$

$$x + 2y - z = 2$$

$$x - y + z = 6$$

Solution

$$2x - 3y + z = 4 \quad \text{--- (I)}$$

$$x + 2y - z = 2 \quad \text{--- (II)}$$

$$x - y + z = 6 \quad \text{--- (III)}$$

Now in matrix form for gauss elimination

$$\left| \begin{array}{ccc|c} 2 & -3 & 1 & 4 \\ 1 & 2 & -1 & 2 \\ 1 & -1 & 1 & 6 \end{array} \right|$$

$$R_2 = R_2 + R_1$$

$$\left| \begin{array}{ccc|c} 2 & -3 & 1 & 4 \\ 3 & -1 & 0 & 6 \\ 1 & -1 & 1 & 6 \end{array} \right|$$

$$R_1 = R_1 - R_2$$

$$\left| \begin{array}{ccc|c} 1 & -2 & 0 & -2 \\ 3 & -1 & 0 & 6 \\ 1 & -1 & 1 & 6 \end{array} \right|$$

Now,

$$x - 2y = -2 \quad \text{--- (IV)}$$

$$3x - y = 6 \quad \text{--- (V)}$$

$$\therefore x = -2 + 2y \quad \text{--- (vi)}$$

Putting the value of x in --- (V) we get

$$3(-2 + 2y) - y = 6$$

$$\therefore -6 + 6y - y = 6$$



$$5y = 6 + 6$$

$$\therefore y = \frac{12}{5}$$

The value of y in — (ii) we get

$$x = -2 + 2y$$

$$\therefore x = -2 + 2 \times \frac{12}{5}$$

$$\therefore x = -2 + \frac{24}{5}$$

$$\therefore x = \frac{14}{5}$$

Putting the value of x, y in — (iii) we get

$$x - y + z = 6$$

$$\frac{14}{5} - \frac{12}{5} + z = 6$$

$$\therefore z = 6 - \frac{2}{5}$$

$$\therefore z = \frac{30 - 2}{5}$$

$$\therefore z = \frac{28}{5}$$

Hence

$$\therefore x = \frac{14}{5}, y = \frac{12}{5} \text{ and } z = \frac{28}{5}$$



② Find the cross and dot product of

$$\vec{a} = 2\hat{i} + 3\hat{j} - 4\hat{k}$$

$$\vec{b} = \hat{i} - 2\hat{j} + 3\hat{k}$$

Solution,

Given,

$$\vec{a} = 2\hat{i} + 3\hat{j} - 4\hat{k}$$

$$\vec{b} = \hat{i} - 2\hat{j} + 3\hat{k}$$

$$\begin{aligned} \text{Dot product } (\vec{a} \cdot \vec{b}) &= 2\hat{i} \cdot 1\hat{i} \cdot \cos 0 + 3\hat{j} \cdot -2\hat{j} \cdot \cos 0 + (-4\hat{k} \cdot 3\hat{k}) \\ &= (2 \times 1) + (3 \times -2) + (-4 \times 3) \\ &= 2 + (-6) + (-12) \\ &= 2 - 6 - 12 \\ &= -16 \end{aligned}$$

$$\text{Cross product } (\vec{a} \times \vec{b}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & -4 \\ 1 & -2 & 3 \end{vmatrix}$$

$$= \hat{i} \begin{vmatrix} 3 & -4 \\ -2 & 3 \end{vmatrix} - \hat{j} \begin{vmatrix} 2 & -4 \\ 1 & 3 \end{vmatrix} + \hat{k} \begin{vmatrix} 2 & 3 \\ 1 & -2 \end{vmatrix}$$

$$\begin{aligned} &= \hat{i} \{ (3 \times 3) - (-4 \times -2) \} - \hat{j} \{ (2 \times 3) - (-4 \times 1) \} + \hat{k} \{ (2 \times -2) - (3 \times 1) \} \\ &= \hat{i} (9 - 8) - \hat{j} (6 + 4) + \hat{k} (-4 - 3) \\ &= \hat{i} - 10\hat{j} - 7\hat{k} \end{aligned}$$



Assignment

4



① Explain Bayes' theorem with examples?

→ Bayes' Theorem is a fundamental concept in probability theory and statistics that describes how to update the probability of a hypothesis based on new evidence.

- It provides a way to calculate conditional probabilities and is widely used in fields like machine learning, data science, and decision-making.

$$\therefore P(H/E) = \frac{P(E/H) \cdot P(H)}{P(E)}$$

where,

- $P(H/E)$ is the probability of hypothesis H being true given the evidence
- $P(E/H)$ the probability of observing the evidence E given that the hypothesis H is true
- $P(H)$ the initial probability of the hypothesis before seeing the evidence
- $P(E)$ is the evidence the total probability of observing the evidence, considering all possible hypotheses.

Example:

Imagine there's a rare disease that 1% of the population has. If you take a test that correctly identifies 95% of people who have the disease, but it also gives a false positive 5% of the time for healthy people. You take the test and get a positive test. What is the probability that you have the disease?



Solution

$$P(H) = 0.01, P(E|H) = 0.95, P(E|\neg H) = 0.05, P(\neg H) = 0.99$$

Using Formula,

$$P(H|E) = \frac{P(E|H) \cdot P(H)}{P(E)}$$

Now

$P(E)$ = total probability of a positive test is

$$\begin{aligned} P(E) &= P(E|H) \cdot P(H) + P(E|\neg H) \cdot P(\neg H) \\ &= (0.95 \times 0.01) + (0.05 \times 0.99) \\ &= 0.059 \end{aligned}$$

Now

$$\begin{aligned} \text{Baye's Theorem} = P(H|E) &= \frac{0.95 \times 0.01}{0.059} \\ &= 0.161 \end{aligned}$$

Hence the probability that you actually have the disease given a positive test result is only about 16.1%.

② If one person draw a ball from a basket which contains 8 red balls, 7 green balls and 5 white ball - what is the probability:

- First ball drawn is of white colour
- Second ball drawn is of red colour
- Third ball drawn is of green colour
- Fourth ball drawn is of white colour
- Fifth ball drawn is of red colour.

one time without replacement and another time with replacement

Solution,

Let's calculate the probability step by step for both with replacement and without replacement cases.

Given,

$$\text{Total Balls: } 8(\text{Red}) + 7(\text{Green}) + 5(\text{White}) = 20$$

Required Sequence:

- First ball = white
- Second ball = Red
- Third ball = Green
- Fourth ball = white
- Fifth ball = Red

Case I (With Replacement)

Since the ball is replaced after each draw, the probability of drawing each colour remains the same.

$$1) P(\text{White first}) = \frac{5}{20}$$

$$2) P(\text{Red second}) = \frac{8}{20}$$

$$3) P(\text{Green Third}) = \frac{7}{20}$$

$$4) P(\text{White fourth}) = \frac{5}{20}$$

$$5) P(\text{Red fifth}) = \frac{8}{20}$$

Since these events are 'independent' (due to replacement) we multiply the probabilities:

$$P = \frac{5}{20} \times \frac{8}{20} \times \frac{7}{20} \times \frac{5}{20} \times \frac{8}{20}$$

$$= \frac{5 \times 8 \times 7 \times 5 \times 8}{20^5}$$

$$= \frac{11200}{320000}$$

$$= 0.035$$

$$= 0.35\%$$

Case 2 (Without replacement)

Since,

The ball is not replaced, the total no. of balls decreases after each draw.

$$1) P(\text{White first}) = \frac{5}{20}$$



$$2) P(\text{Red Second}) = \frac{8}{19} \text{ (One ball already drawn)}$$

$$3) P(\text{Green Third}) = \frac{7}{18}$$

$$4) P(\text{White Fourth}) = \frac{4}{17} \text{ (One white ball already drawn)}$$

$$5) P(\text{Red Fifth}) = \frac{7}{16} \text{ (One red ball is already drawn)}$$

Ans

Multiplying the probability

$$P = \frac{5}{20} \times \frac{8}{19} \times \frac{7}{18} \times \frac{4}{17} \times \frac{7}{16}$$

$$= \frac{7840}{1860480}$$

$$= 0.0042$$

$$= 0.42\%$$

Result:

with replacement = 0.35% or (0.0035)

without replacement = 0.42% or (0.0042) ~~#~~